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PRACTICAL REPORT: MOLECULAR MODELLING

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Molecular Dynamics Simulations with BRD4

Introduction

BRD4 is a bromodomain-containing protein involved in the regulation of gene expression through recognition of acetylated lysine residues on histone proteins. Because of its role in transcriptional regulation and its implication in several diseases, including cancer, BRD4 has become an important target for drug development. Understanding how ligands interact with the BRD4 binding pocket and how mutations affect these interactions is essential for studying protein function and ligand stability.

Molecular modeling and molecular dynamics (MD) simulations allow the investigation of protein structures and interactions at the atomic level. They provide information about how proteins and ligands move over time, making it possible to study structural flexibility, binding stability, and the effects of mutations on protein dynamics. In this project, these modelling methods were used to analyze the structural behaviour of BRD4 in different conditions and to determine how two point mutations influence both the protein and its ligand.

The main goal of this practical was to evaluate the effects of the point mutations in the binding pocket on the structural dynamics of BRD4 and on the behavior of the LSH24 (segment HETB) ligand (only one of the two copies in the crystal structure was simulated). We aimed to determine whether the mutations alter the stability of the protein and whether they influence the ligand within the binding site.

Methods

We analyzed three types of the protein (PDB 7AXR [1]): the Apo form containing only the chain A and no ligand, the Holo form consisting of BRD4 bound to the ligand LSH24 (resname S7T, segment HETB), and a mutant form also containing the ligand and in which two amino acids were mutated (Trp81 and Phe83). These amino acids are located near the binding pocket and are likely involved in stabilizing ligand interactions through hydrophobic or aromatic contacts. Replacing them with alanine reduces side-chain size and interaction potential, which may affect binding pocket geometry, protein stability, and ligand positioning.

The structures were prepared using CHARMM-GUI [2], with the CHARMM36m [3] protein force field and CGenFF [4] ligand parameters. The simulations were performed with NAMD [5] and visualized and analyzed in VMD [6]. Energy minimization was first applied to all protein structures for 10,000 steps to relax steric clashes and reach a local energy minimum; a plateau in the root mean square deviation (RMSD) shows when the structure stopped changing. Molecular dynamics simulations in vacuum, with no thermostat (1 fs steps, 200 ps in total), were then run to generate trajectories for further analysis.

Results

Identification of point mutations in the binding pocket of BRD4

PyMOL was used to compare the mutant amino acid sequence to the WT sequence. By comparing the alignment, two point mutations were found in the binding pocket of the protein *(Figure 1):*

1. Trp81→Ala
2. Phe83→Ala

Both tryptophan and phenylalanine are aromatic residues that can contribute to hydrophobic interactions and stabilization of ligand binding. Their replacement with alanine reduces side-chain size and hydrophobic surface area, which may influence the local environment of the binding pocket. Because of this, the effect of these point mutations on both the protein structure and the ligand behavior was analyzed in the following steps.

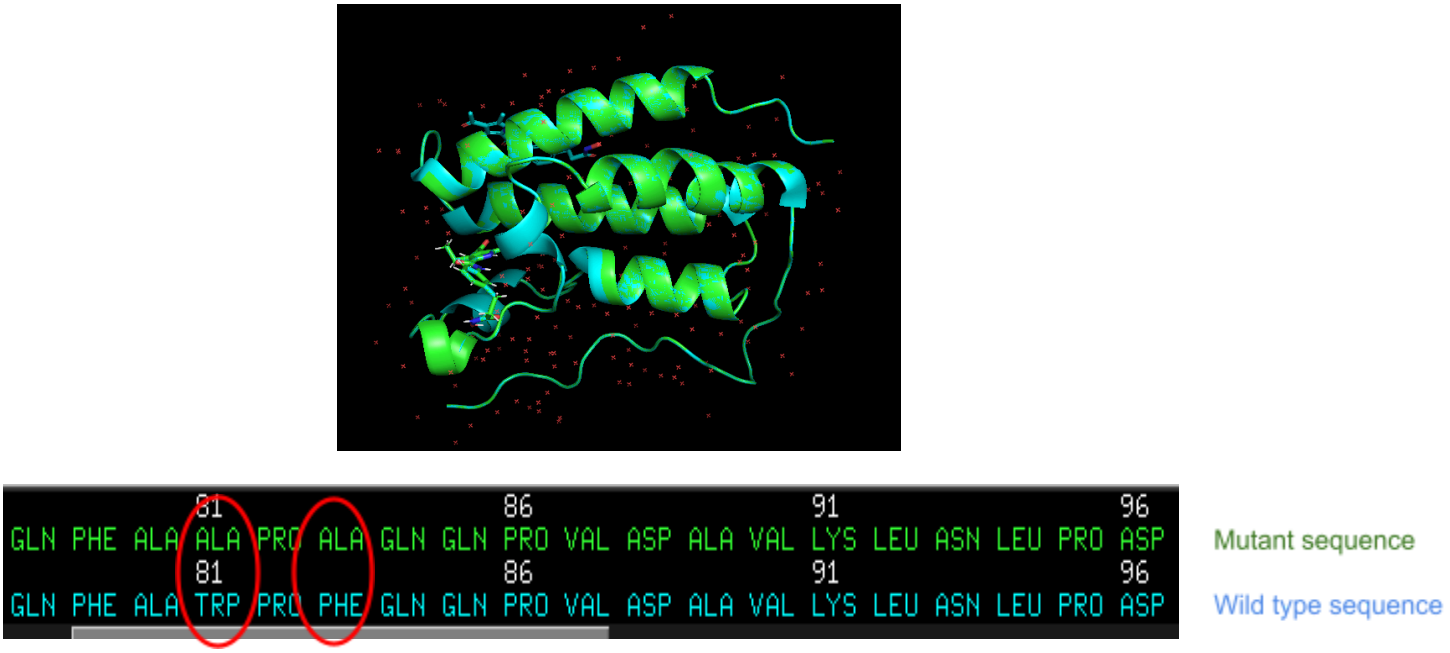


Figure 1: Identification of mutated residues in BRD4. *Top:* PyMOL superposition of the wild-type and mutant structures. *Bottom:* sequence alignment of residues 78–96 (mutant in green, wild type in cyan); the circled Trp81→Ala and Phe83→Ala substitutions are the only differences.

Energy minimization

Before performing molecular dynamics (MD) simulations, energy minimization was done for all protein systems to get an energetically stable conformation. NAMD minimization was performed with 10,000 steps, after which the RMSD profile reached a plateau, indicating that the structure had stopped changing appreciably (Figure 2).

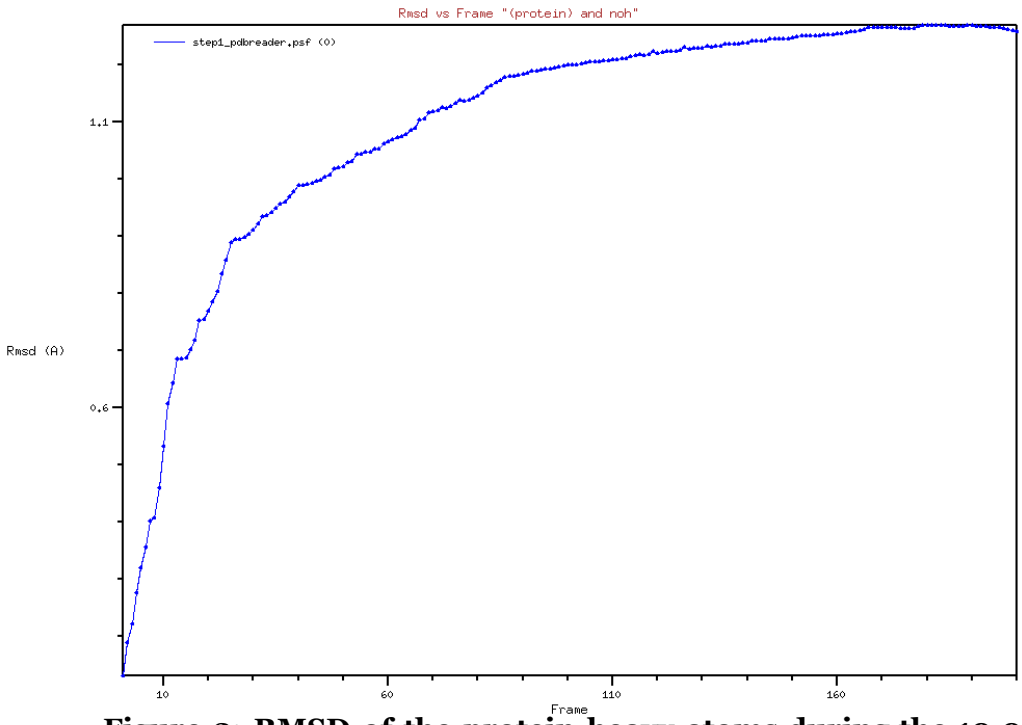


Figure 2: RMSD of the protein heavy atoms during the 10,000-step minimization of the mutant system (one frame every 50 steps). The curve flattens at about 1.2 Å.

Apo vs holo

After the minimization, Molecular Dynamics simulations were done for the Apo, Holo, and mutant proteins with 200,000 steps. The resulting trajectories were aligned and compared: we observe firstly that the apo reaches an RMSD around 2.5 Å and stays there whereas the holo stays at ~1.3–1.5 Å, and ligand binding is associated with a more stable fold in this single vacuum run (Figure 3).

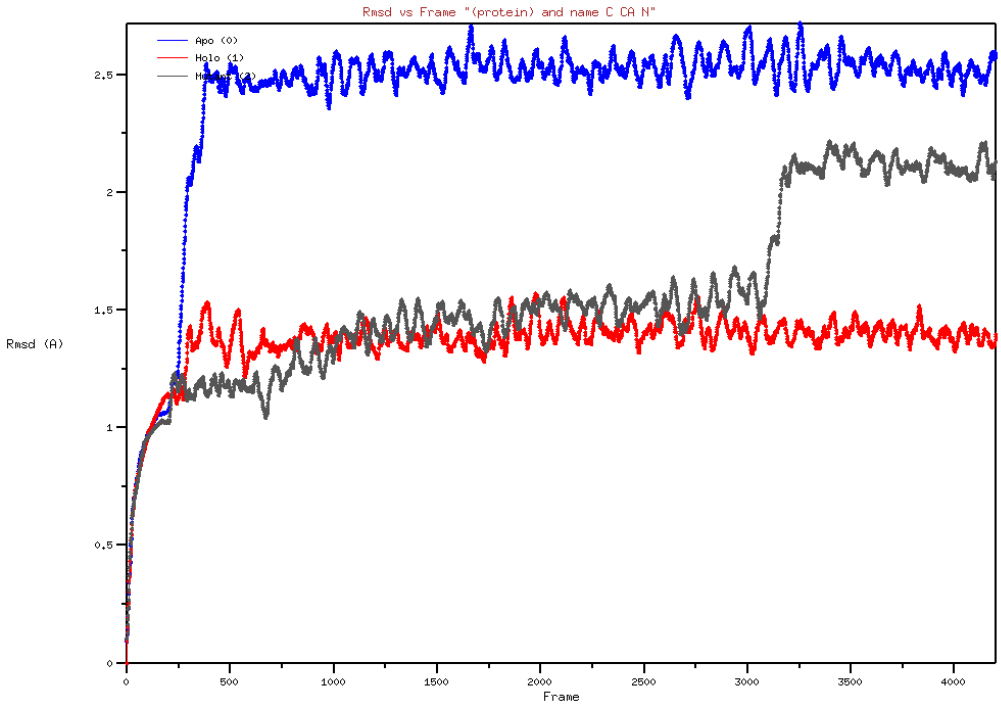


Figure 3: RMSD analysis of protein structural dynamics.
Backbone (N, CA, C) RMSD of the apo (blue), holo (red) and mutant (gray) systems. One frame is 0.05 ps and the first ≈200 frames are minimization, so frame 3100 ≈ 145 ps of MD. Apo plateaus at ≈2.5 Å and holo at ≈1.3–1.5 Å; the mutant follows holo until ≈145 ps, then rises to ≈2.1 Å.

Effect of the mutations on the protein

As for the mutant protein, it displayed a change in dynamics (from about 1.5 Å to 2 Å) between ≈145–152 ps (visible in Figure 3). This shift corresponds to a conformational change that was also visible in the visualization tool (Figure 4). These observations align with localized flexibility or structural rearrangement within the protein; however, a single simulation trajectory is insufficient to definitively link this behavior to the point mutations.

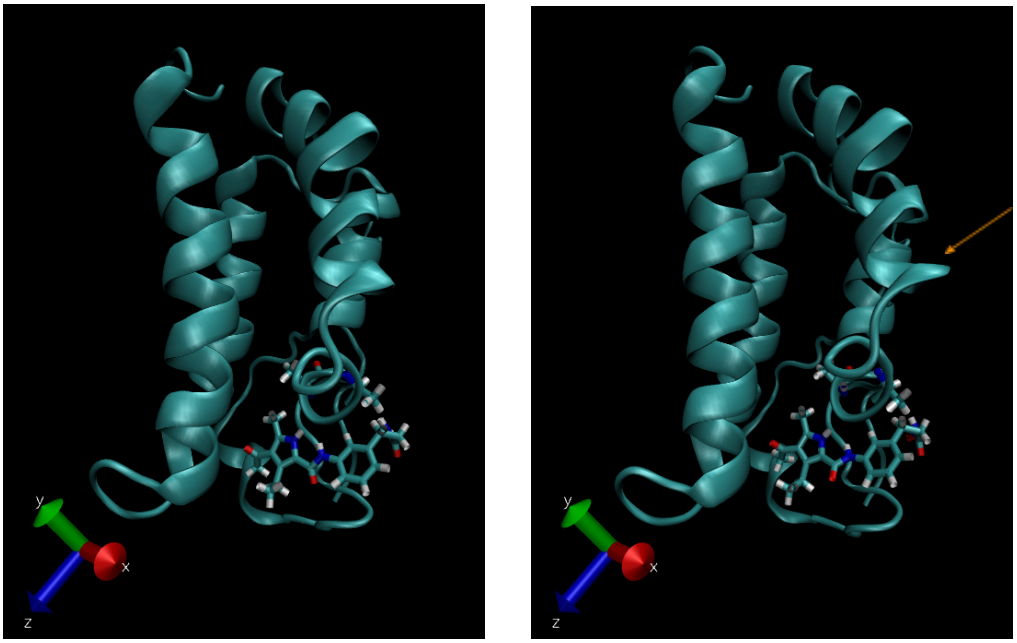


Figure 4: Ligand-bound mutant protein at different simulation frames. *Left:* structure at frame ≈3100 (≈145 ps). *Right:* structure at the end of the trajectory (frame ≈4200, ≈200 ps), showing a conformational change (highlighted by an orange arrow).

Effect of the mutations on the ligand

Despite the observed conformational differences between the mutant and holo protein structures, no obvious difference was detected in the RMSD of the ligand when comparing the holo and mutant proteins. The RMSD plot (Figure 5) and visualisation (Figure 6) show similar behavior for both types of protein, indicating that the ligand remains relatively stable within the binding pocket during the simulation. This suggests that the two point mutations do not seem to affect ligand positioning, and that the ligand stays within the binding pocket in both systems.

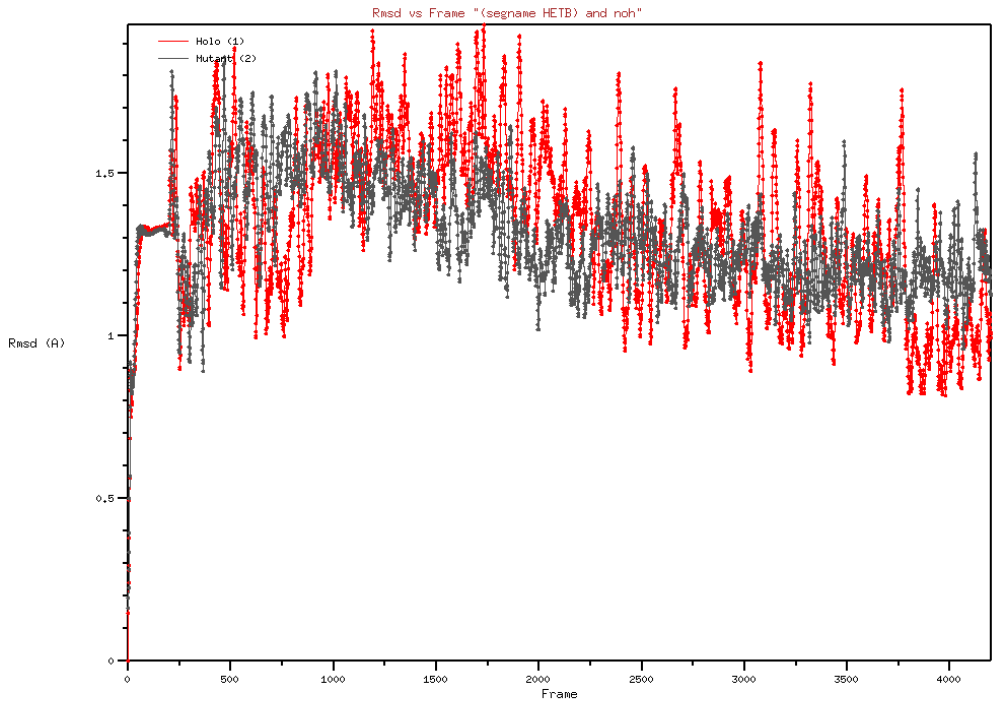


Figure 5: RMSD comparison of ligand–protein dynamics.
RMSD of the LSH24 heavy atoms (segment HETB) in the holo (red) and mutant (gray) systems; both stay between ≈0.9 and 1.9 Å.

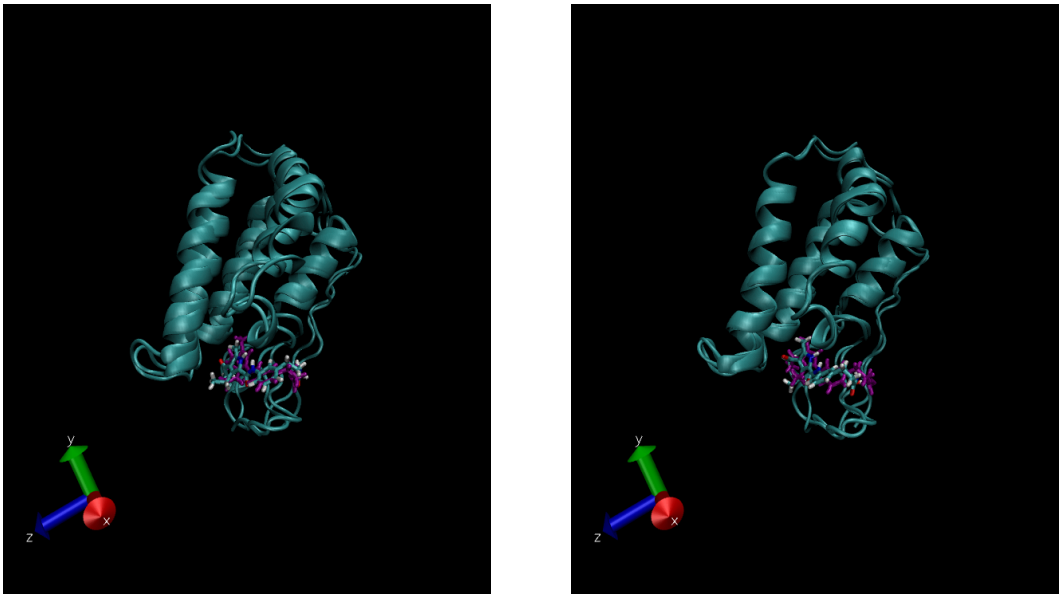


Figure 6: Visualization of the aligned holo and mutated protein BRD4. Overlay of the holo and mutant systems at the start (*Left*) and end (*Right*) of the trajectory. The ligand remains in the binding pocket in both.

Discussion

The replacement of Trp81 and Phe83 with alanine was expected to alter the hydrophobic environment of the binding pocket because aromatic residues often play important roles in stabilizing ligand interactions. This 200 ps vacuum simulation indicates that while the mutations introduce detectable changes in protein dynamics, these changes appear to be localized. Despite these structural differences, the ligand RMSD analysis shows that the overall binding and the ligand itself remain stable. This may indicate that the mutations at positions 81 and 83 are not sufficient to have an effect on the ligand within 200 ps. The apo system also drifted further from its starting structure than the holo system, which suggests that the ligand stabilizes the bromodomain in this simulation.

Limitations

- The simulation length may influence the observed results. Molecular dynamics simulations capture only a limited timescale, and longer simulations could potentially show ligand movements that are not visible within the current trajectory (200 ps).
- We have only one replica per system, whereas MD is stochastic and needs replicas to account for run-to-run differences.

- We worked in a vacuum with no thermostat (temperature not controlled), not in a typical protein environment.
- Poor analogy in CGenFF ligand parameters: one torsion has a penalty of 68 (above 50).
- RMSD describes positional stability, not binding affinity, and no free-energy estimate was made.

Conclusion

In this single 200 ps vacuum simulation, the ligand-bound BRD4 bromodomain was more stable than the apo form. Replacing Trp81 and Phe83 with alanine was followed by a localized conformational change in the protein but no obvious change in the position of LSH24 in the binding pocket. Replicas, longer simulations and a solvated, temperature-controlled setup would be needed to confirm these observations.

References

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